

Web TV Design Document

Emma Kilcoyne

Overview

Summary

The Web TV app will be a desktop application that organizes media files. Users can select a folder of movies, music, audiobooks, ebooks, then create categories to organize the files and then view all these files within the app. Making it a one stop shop for watching videos, listening to music, and reading.

Goal

The goal of this project is to be able to create a user interface that easily allows users to organize and view their media files. This app will contain a media player for videos, audio, and a document viewer.

Functional Concept

Below are images representing each of the windows of the application as well as any buttons. These “functional concept” images are scrapped of aesthetic design for the purpose of showcasing the bones of the app.

Home Window

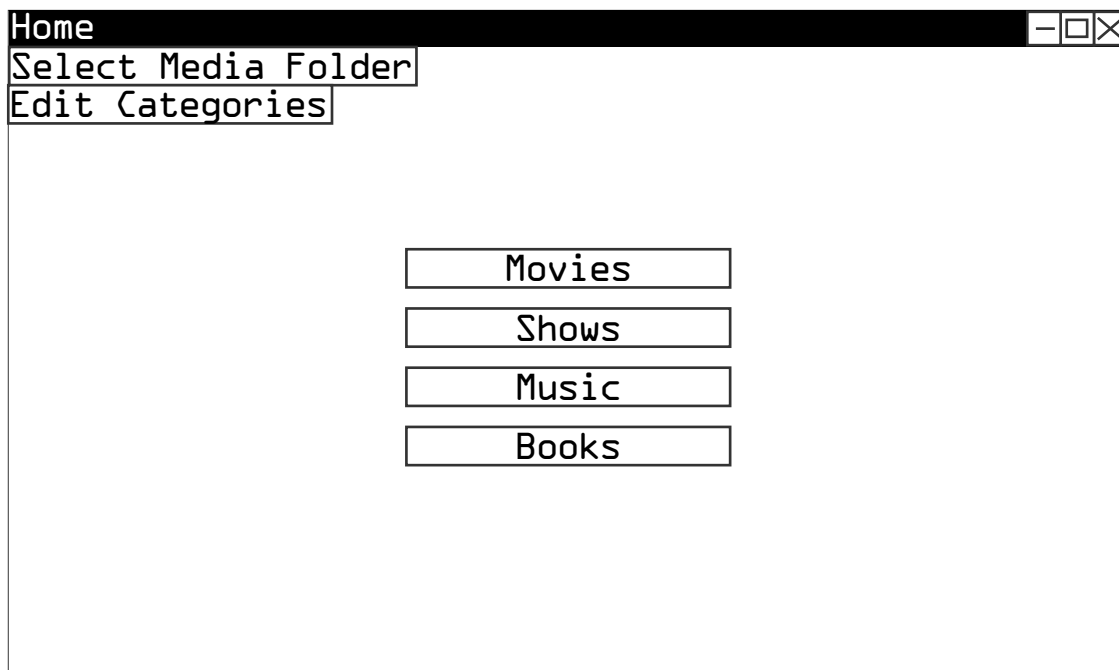
- Select Media Folder

This allows the user to select the folder on their local device that contains their media.

- Edit Categories

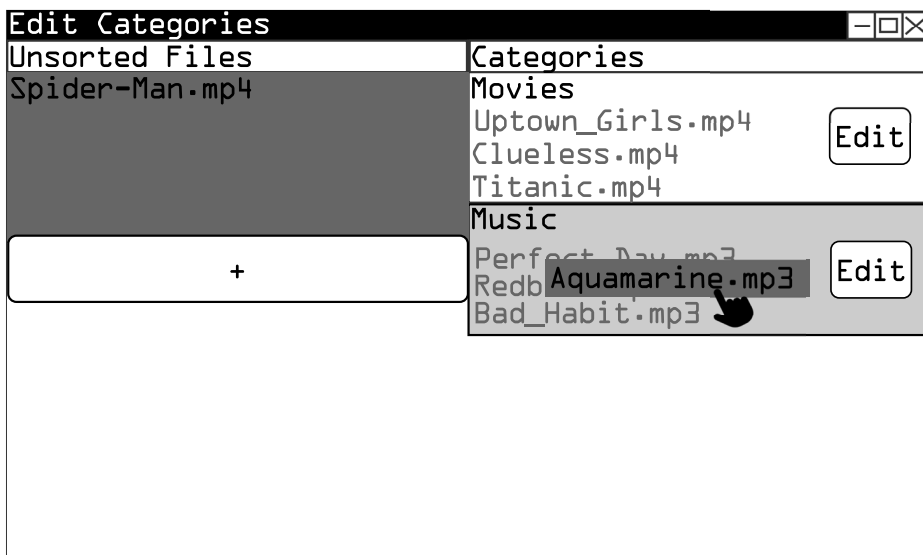
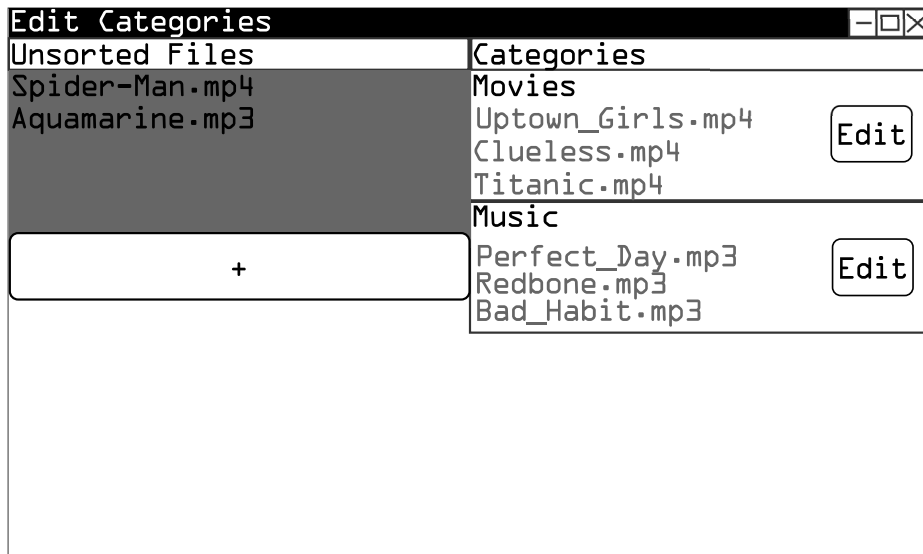
Opens the Edit Categories Window

- Displays Categories with links to their respective directory



Edit Categories

- Users can drag and drop unsorted files to the designated category
- Delete existing categories such as the default categories [Movies, Music, Books]
- Create new categories such as [Meditations, Workout Videos, Manga]



Edit Categories	
Unsorted Files	Categories
Spider-Man.mp4	Movies Uptown_Girls.mp4 Clueless.mp4 Titanic.mp4 Music Perfect_Day.mp3 Redbone.mp3 Bad_Habit.mp3 BooksI
+	Edit Edit Edit Edit

[Insert name of category] Directory ie. (Movie Directory)

- List of Media Files in that category
- Each is a link to the respective player ie. (.mp4 file directs to the Video Media Player window)

- Video Media Player
- plays video content
- Audio Media Player
- plays audio content
- Document Viewer
- viewer for document or image files

UX/UI

Visual Design Inspiration

Windows XP's builtin Media Player.

“Frutiger Aero”

Glass UI / Glossy 3D Buttons

User Experience

The user selects a local folder containing media files that they would like to see in app.

The user then can create/delete categories and drag and drop each file into their respective categories

Each Category will have it's own window (directory)

The media in that category will be in a list on that page

UI Concept

Finalized Aesthetic concept for each window

- Assets : Buttons, Icons, Font, Backgrounds, Borders

Technical Design

Technology Stack

- HTML
- CSS
- Javascript
- Electron
- SQLite

Assets will be built in Illustrator or Photoshop and imported into CSS. Functions will be done in Javascript. The app interface will be done in Electron and the app would be a web-based desktop app.

Javascript will ask the user for access to a specific media file then from that file two tables will be created in SQLite. One table with file information, file name, file type, size. The next table would be for categories.

Electron will be used for the interface.

Each time a file is selected from a directory it will need to be sent to its corresponding media player or viewer.

Frontend

The application interface will be built using HTML, CSS, and JavaScript within Electron.

Visual assets such as icons, buttons, and interface graphics will be created in Adobe Illustrator and Photoshop, then exported and integrated into the application through CSS and HTML.

JavaScript will handle user interactions, including folder selection, media browsing, category management, and communication with the database.

Database

SQLite will store metadata about the scanned media files in the user selected media folder.

There will be two tables: one for file metadata, and another for categories.

File Table

Stores information about each imported file

Fields:

- file_name
- file_type
- file_size
- file_path

Categories Table

Stores the user defined categories.

Fields:

- category_name
- file_name
- file_type
- file_path

Backend Logic

The user selects a directory containing media files. The application scans the selected directory and extracts metadata from each file. For every discovered file, a record is created in the File table containing information such as file name, type, path and size. Users may assign categories to media files.

Media Viewing Workflow

1. User selects a media file from the categories interface.
2. The application retrieves the file path from the SQLite database.
3. JavaScript determines the appropriate viewer based on the file type.
4. The selected file path is passed to the corresponding media viewer component.
5. The viewer loads and displays the media.

Data Flow

1. User Selects Folder
2. Application Scans Files
3. Metadata Extracted
4. SQLite Database Updated
5. User creates categories
6. User drags and drops files into categories
7. Category table is updated with file information
8. User Browses Category Directory
9. User Selects Media File
10. File Path Retrieved
11. Media Viewer Opens Selected File

Future Updates

Smart Sort

Smart Sort would be a way for the program to suggest what category an unsorted file should go to based on patterns of previously added files. For example, if in the movies category there are only .mp4 files that are more than 45 minutes in length but less than 3 hours, then the smart sort would suggest that added files with that same pattern go into the movies category. The user would select smart sort on the unsorted files and have the opportunity to change and move them before committing to the changes.

Add multiple folders

When selecting the folder that media is in, the ability to add multiple file locations would be a nice addition.

Add Themes

Themes would be able to be added to customize the look of each category or playlist. The user would be able to change the color associated with a specific category and/or upload an image to act as a cover. Perhaps in a later version there would be a set of themes to choose from that include different UI packets or even the ability for the user to customize their own UI.